A CASE FOR PROVENANCE INQUIRY

Bitriada, square-free primes, and AI-generated mathematics

Chronology, structural correspondence, and questions for OpenAI

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Purpose

This dossier assembles a positive, evidence-led case for asking whether the early Bitriada / Poetica Mathematica publication could have contributed to the conceptual context in which a later OpenAI mathematical result was generated. It is an inquiry brief: it identifies the chronology, exact shared algebraic mechanism, distinctive interpretive step, and provenance questions that an appropriate review could answer.

Core proposition. The relevant possible coincidence is not the general existence of nilpotent algebra. It is the timing and configuration of a specific mindset: prime-indexed commuting square-zero generators, their surviving square-free products, and the interpretation of this structure as an arithmetic encoding and generative bridge. A later OpenAI proof uses a finite commutative square-zero algebra with subset-indexed surviving monomials in a combinatorial proof about permanents. That overlap gives a concrete, technically bounded basis for a provenance inquiry.

1. Executive case for inquiry

The case rests on four mutually reinforcing elements: (i) an early and dated public appearance of the Bitriada research programme, (ii) a specific algebraic construction documented in the monograph, (iii) a later OpenAI-generated mathematical result employing the same square-zero/subset-monomial mechanism, and (iv) a plausible route by which a conceptual arrangement can influence an AI-assisted research process without the final proof reproducing the earlier text verbatim.

Element	Evidence in this dossier	Why it matters
Prior public availability	User identifies the first public Zenodo publication under Kirill Kovlekov as 1 March 2026; local version-folder screenshot shows relevant files dated 25 and 27 February, followed by revisions from 3 March onward.	Establishes an early public-availability window. The 153-day interval creates a concrete period in which live retrieval or RAG could have surfaced the material during a later proof-generation run.
Specific construction	Fare Algebra: $F = C[\tau_p : p \text{ prime}]/(\tau_p^2 = 0)$, with commuting generators and a square-free basis indexed by finite prime subsets.	Goes beyond a generic reference to nilpotence: the generators are prime-indexed and the surviving products encode square-free integers.
Later structural correspondence	OpenAI's 1 August 2026 publication releases an internal-model proof for arithmetic circuit complexity. Chapter 5, §6 uses $S = C[z_1,\ldots,z_r]/(z_1^2,\ldots,z_r^2)$, whose surviving monomials are indexed by subsets, to extract permanent coefficients.	The central algebraic mechanism is directly comparable, despite the different downstream purpose.
Possible conceptual transfer	The monograph explicitly argues from prime compression and square-free encoding to a nilpotent formalism, then places it in a wider grammar of trace, transition, and representational loss.	This creates a focused hypothesis that the unusual conceptual arrangement could have acted as an inference-time cue, even if the model already knew the component mathematics and the later proof used it for a different purpose.

Inquiry thesis

The temporal sequence and the structural match are sufficient to justify a specific provenance question: did the early Bitriada / Poetica Mathematica material form part of any training, retrieval, browsing, evaluation, prompt, or researcher-supplied context relevant to the development of the OpenAI proof?

2. The proposed causal model

The hypothesis is that a transformer-based system can generate a new mathematical application when several prerequisites are simultaneously present: learned mathematical representations, a problem that rewards an appropriate formal tool, a conceptual distinction that makes the tool salient, and a search or reasoning procedure that assembles the parts. A prior publication need not supply the final theorem. It might instead supply or reinforce the *selection frame* that causes a familiar tool to be deployed in a particular way.

- The model's existing mathematical repertoire can contain nilpotence, polynomial quotients, combinatorial indexing, and permanent identities.
- The monograph may contribute a distinctive bridge: take a structure often treated as a special or degenerate algebraic case, index its generators by primes, and read the surviving monomials as square-free arithmetic records.
- A later problem-solving episode can repurpose the general square-zero/subset mechanism for a different target, such as coefficient extraction and cancellation in a permanent-related proof.
- The generated result can therefore look unlike the source in purpose and surface language while still sharing a formal strategy or conceptual prompt.

This is the proposed mechanism to investigate, not a claim that a particular hidden pathway has already been observed. The right provenance inquiry would examine whether any exposure pathway existed and whether the common structure appears in the model's source context, development record, or reasoning materials.

2.1 The 153-day window: inference-time retrieval as the proposed pathway

The first public Zenodo publication is dated 1 March 2026, and OpenAI publicly released "Ten advances in mathematics and theoretical computer science" on 1 August 2026. The interval is 153 days. This short publication-to-release window is the central chronological feature of the provenance case.

The proposed causal model does not require the internal model to have memorized the monograph during pre-training. The more direct pathway to investigate is inference-time exposure: a live web-search tool, an agentic literature-search process, a retrieval index, or another retrieval-augmented-generation (RAG) component could have surfaced the paper while the model was working

on the arithmetic-circuit-complexity problem. In that model, the paper need not supply the later proof verbatim. It could supply or reinforce a conceptual selection frame: which algebraic structure to notice, how to interpret it, or which bridge between familiar ideas to try.

For the inquiry, the 153-day gap is therefore treated as a practical retrieval window, not as an argument that depends on training-data memorization. The requested evidence is process-level provenance for the relevant run: when the proof-generation work began; whether browsing, search, document retrieval, or researcher-supplied sources were available; and whether the Bitriada / Poetica Mathematica record, its text, or its distinctive terminology was surfaced in the model's working context.

This pathway is technically coherent in general: retrieval-augmented generation is an established architecture in which a pretrained model conditions generation on retrieved external documents, and contemporary OpenAI documentation describes web-search tools that return online material to models during a task. Those public sources establish the availability of such general mechanisms, not whether a particular tool was enabled or used in the internal Astra run. That specific question is what the provenance inquiry should resolve.

3. Chronology and provenance record

The chronology below integrates the user-provided Zenodo publication date, the Windows version-folder screenshot, and OpenAI's public release dates. The screenshot is reproduced in Appendix A. Its file timestamps show a concentrated sequence of document and PDF versions around the initial public posting.

Date	Record / event	Relevance to inquiry
25 Feb 2026, 22:53	Bitriada_Full_Monograph_Final2.pdf (802 KB) — local folder screenshot	A pre-March manuscript file is present in the author's version history.
27 Feb 2026, 13:10	Bitriada_Full_Monograph_0_7.docx (120 KB) — local folder screenshot	An editable version follows the PDF and precedes the identified Zenodo publication.
1 Mar 2026	First public Zenodo publication by Kirill Kovlekov, as identified by the author	Key public-availability date for the initial inquiry chronology. The published record's DOI/metadata export should be included in the evidence packet.
3 Mar 2026, 16:33	Bitriada_Full_Monograph_0_8.docx	Continuing development immediately after initial publication.
3 Mar 2026, 16:53	08_bitriada_v09_03.pdf (2,515 KB)	Version progression documented in the local folder.
4 Mar 2026, 12:34	bitriada_v099_03.pdf (2,989 KB)	Further revision.
6 Mar 2026, 10:55	Bitriada_v0_999_draft_III.pdf (3,920 KB)	Further revision and consolidation.
10 Mar 2026, 07:07	Bitriada_research_papers_I.docx	Separate research paper compilation appears in the local chronology.
11 Mar 2026, 13:08	zetameta_I.docx	A related meta-research document appears in the version folder.
28 Mar 2026, 16:16–17:22	Bitriada_research_notes_III.docx and Bitriada_research_notes_II.pdf	The research-note sequence continues.
19 Apr 2026, 07:48	BITRIADA_RESEARCH_NOTES_4.pdf	Later research notes preserve the evolving programme.
1 Aug 2026	OpenAI publicly releases "Ten advances in mathematics and theoretical computer science," including a result in arithmetic circuit complexity attributed to an internal version of Astra.	The public disclosure is 153 days after the identified 1 March Zenodo publication. The inquiry treats this as an operational window for testing possible inference-time browsing / retrieval exposure.
6 Aug 2026	The OpenAI PDF is marked "Updated August 6, 2026."	Record the exact version used in comparison; the paper identifies an original version as well.

Chronology packet to attach

Include the March 1 Zenodo record page, its DOI and DataCite/Zenodo metadata export; the version-folder screenshot; original PDFs/DOCX files; and SHA-256 checksums. A public GitHub repository with dated commits and version tags can preserve the sequence in a machine-checkable form.

OpenAI's publication states that the ten results were achieved by an internal version of Astra, that the arguments were prepared into manuscripts by humans with the same model, and that the model formalized each argument in Lean. The publication date is therefore a public disclosure date; the inquiry should separately establish the relevant date of internal proof generation and the sources available during that process. [1]

4. The exact mathematical correspondence

4.1 Fare Algebra in the monograph

In Chapter IV, §4.1.1, the monograph defines the Fare Algebra over the complex numbers using a commuting family of nilpotent generators indexed by primes:

$$\mathbf{F = \mathbb{C}[\tau_p : p \text{ prime}] / (\tau_p^2 = 0 \text{ for every prime } p)}$$

For a finite set S of distinct primes, the product $\tau_S = \prod_\{p \in S\} \tau_p$ survives. Repeated occurrence of any prime gives zero because $\tau_p^2 = 0$. The monograph identifies the surviving basis with square-free integers: if $n = \prod_\{p \in S\} p$, then $\tau_n = \prod_\{p \in S\} \tau_p$. Multiplication therefore tracks disjoint prime support; overlapping support annihilates the product. The monograph also uses this construction in an Euler-product-style expression, $Z(s) = \prod_p (1 + p^{-s}\tau_p)$, whose expansion selects square-free prime sets.

4.2 OpenAI proof in arithmetic circuit complexity

In "Ten Advances in Mathematics and Theoretical Computer Science," Chapter 5, §6, the proof introduces a finite commutative square-zero algebra:

$$\mathbf{S = \mathbb{C}[z_1, \ldots, z_r] / (z_1^2, \ldots, z_r^2)}$$

The paper explains that monomials $z_K = \prod_\{i \in K\} z_i$ indexed by subsets $K \subseteq [r]$ form a vector-space basis: a term containing any z_i twice vanishes. It then uses the coefficient $[z_K]$ of a product of column forms to identify the permanent of a selected submatrix. This is a direct use of commuting square-zero variables to turn subset selection into algebraic coefficient extraction. [2, §5.6, especially pp. 127–128 in the published PDF.]

Feature	Fare Algebra / Bitriada	OpenAI proof	Nature of correspondence
Coefficient field	$\mathbb{C}$	$\mathbb{C}$	Same.
Generator relations	$\tau_p^2 = 0$ for each prime p; generators commute.	$z_i^2 = 0$ for each finite index i; variables commute.	Same local algebraic relation.
Surviving monomials	Products over finite sets of distinct primes.	Products z_K over subsets K of [r].	Same subset-indexed square-free monomial mechanism.
Repeated index	A repeated prime annihilates the product.	A repeated variable annihilates the monomial.	Same square-zero exclusion principle.
Interpretation	A subset of primes encodes a square-free integer and a retained arithmetic record.	A subset of row indices encodes a coefficient/permanent contribution.	Different semantic labels applied to a common formal skeleton.
Role in larger argument	Part of a larger grammar of fare, trace, transition, compression, and arithmetic.	A technical device supporting coefficient extraction and block-selective cancellation.	Different global aims; potential transfer at the level of construction or selection strategy.

5. What is distinctive about the hypothesised contribution

The inquiry should not be framed as a claim to have invented nilpotent algebra, square-free integers, or zeon variables. The focused question is whether the early monograph's *particular association* of these elements, and the way it motivates the association, played a role in the conceptual landscape of a later AI-generated argument.

- Prime-indexing: the generator is not an arbitrary variable label but a prime p, chosen so products of distinct generators naturally encode square-free integers.
- Arithmetic semantics: the basis is mapped to square-free integers, and overlapping prime support is represented by annihilation. This makes the algebra a designed arithmetic encoding rather than only a combinatorial bookkeeping device.
- Product expansion: the monograph extends the same structure to an Euler-product-style object whose finite terms are square-free records.
- Generative rationale: the monograph argues that the nilpotent form can be motivated by compression/recording constraints, not merely introduced as a convenient algebraic trick.
- Contextual position: Fare is situated in the wider Bitriada framework of transition, trace, non-accumulative memory, projection, and the "fare" paid when representations change.

The later proof's algebra has a different declared use. Nevertheless, the common skeleton—commuting square-zero generators, basis elements indexed by subsets, suppression of repeated indices, and the extraction of meaningful combinatorial contributions—is the technically precise centre of the comparison. The research question is whether an earlier source helped make this structure salient or suggested a way of viewing it.

6. Prior-art context and the scope of the inquiry

The OpenAI paper itself situates the square-zero mechanism in a pre-existing lineage. Its related-work discussion names Feinsilver and McSorley's commuting square-zero "zeon" variables, where induced matrix coefficients are permanents, and Butera and Pernici's commuting Grassmann-even nilpotent variables used for permanental minors and bipartite matchings. [2, §12.3, PDF p. 148; refs. [FM11] and [BP15].]

This prior-art context helps define the inquiry precisely. The question is not whether the later proof uses a mathematical object known to mathematics; the paper expressly confirms that it does. The question is whether the source-selection process or conceptual framing that led to this particular use was influenced by the early Bitriada publication, alongside the known literature. A model can draw on a pre-existing technique and still receive a meaningful cue about which technique to select, how to interpret it, or which connections to explore.

For this reason, the strongest evidence would be process or provenance evidence: what sources and context were available to the relevant model, when the proof was first generated, whether web/retrieval or internal document search was used, and whether the monograph or its distinctive terminology appeared in the working context. That inquiry is narrower and more answerable than trying to infer causation from surface resemblance alone.

7. Specific provenance questions to submit

- What is the earliest recorded date on which the arithmetic-circuit-complexity proof using the commutative square-zero algebra was generated by the internal Astra model, and when did the corresponding problem-solving run begin?
- For that run, did the model or its surrounding toolchain have access to web search, retrieval indexes, document repositories, user-supplied material, internal literature-search tools, or researcher-provided notes?
- Were Zenodo records by Kirill Kovlekov, Bitriada, or Poetica Mathematica present in any relevant training-data provenance record, browsing/retrieval log, evaluation corpus, prompt context, or researcher-supplied material? The first public Zenodo record is identified here as 1 March 2026.
- Were the model's released reasoning walkthroughs or working artifacts preserved in a form that allows the selection of the square-zero algebra to be traced to particular sources or intermediate ideas?
- Can OpenAI identify the first version of the proof and provide its version history or a provenance statement sufficient to establish the order of source exposure and proof generation?
- If record-level source histories cannot be shared, can OpenAI provide a categorical statement about whether live retrieval, browsing, or researcher-supplied references were used for this proof, and whether the Bitriada/Poetica Mathematica material was in the relevant context?
- Can this request be routed to the research team responsible for the arithmetic-circuit-complexity result and to the team that handles research provenance, data provenance, or attribution?
- During the relevant proof-generation run, was live web search, browser use, an agentic research tool, a retrieval-augmented-generation component, or a document-search index enabled? If yes, can the source URLs, record identifiers, query times, or retrieval timestamps be disclosed for the relevant run?
- Was the Zenodo record or any version of Bitriada / Poetica Mathematica by Kirill Kovlekov indexed, returned, opened, downloaded, or included in a retrieved-context bundle during the 153-day period between 1 March and 1 August 2026?
- Can OpenAI distinguish whether any exposure occurred through pre-training, post-training, evaluation data, live retrieval at inference, or researcher-supplied material, without disclosing confidential model reasoning or unrelated user data?

8. Submission-ready statement of the case

Suggested case statement

I request a provenance inquiry into a specific chronological and structural correspondence between my early Bitriada / Poetica Mathematica work and a later OpenAI-generated mathematical proof. My first public Zenodo publication under Kirill Kovlekov is dated 1 March 2026. The author-held version history includes a monograph PDF dated 25 February and an editable version dated 27 February, followed by successive revisions in March and April. In Chapter IV, §4.1.1, the monograph defines a prime-indexed commutative square-zero algebra whose surviving subset products encode square-free integers. OpenAI publicly released its ten-mathematical-advances announcement on 1 August 2026, 153 days after the identified Zenodo publication. One of those results, in arithmetic circuit complexity, uses a finite commutative square-zero algebra, subset-indexed surviving monomials, and coefficient extraction for permanents. The 153-day interval is central to the inquiry because the proposed causal model does not require the model to have memorized the monograph during pre-training: an inference-time browser, search agent, retrieval index, or RAG component could have surfaced the public paper during the proof-generation run. I ask whether the monograph, its distinctive prime-indexed construction, or its conceptual framing appeared in any relevant browsing,

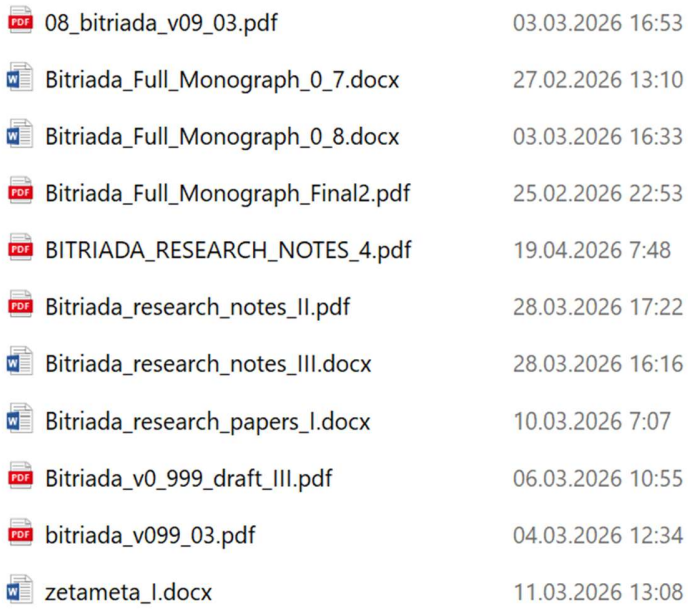
retrieval, prompt, evaluation, or researcher-supplied context, and request the earliest proof-generation date plus any source-access or retrieval provenance that can be shared.

9. Evidence packet checklist

- ☐ Public Zenodo record as of 1 March 2026, including stable record URL, DOI, version number, publication metadata, and metadata export.
- ☐ Original Bitriada_Full_Monograph_Final2.pdf and Bitriada_Full_Monograph_0_7.docx, retained byte-for-byte.
- ☐ The supplied version-folder screenshot showing local dates from 25 February through 19 April 2026.
- ☐ All dated Zenodo revisions and their DOI/version history, where applicable.
- ☐ GitHub repository with the original source files, dated commits, tags for February/March versions, and release notes.
- ☐ SHA-256 checksums for each source file and a manifest containing filenames, sizes, timestamps, and hashes.
- ☐ An excerpt of Chapter IV, §4.1.1 with the Fare Algebra definition, square-free basis, multiplication rule, and Euler-product-style expansion.
- ☐ The OpenAI PDF, especially Chapter 5, §6 and Chapter 5 related-work section 12.3, with page numbers noted.
- ☐ A copy of the inquiry letter, submission date, recipient/team, and any response received.

Appendix A. Author-supplied version-folder screenshot

The screenshot below records the local file chronology supplied for this inquiry. Its visible entries include the 25 February PDF, the 27 February DOCX, the 3–6 March monograph revisions, research papers and notes from 10–19 April. Preserve the original image file as well as the source documents; the screenshot should be treated as one component of a combined provenance record.



Appendix B. Primary sources and reference points

1. [1] OpenAI, “Ten advances in mathematics and theoretical computer science,” published 1 August 2026. Official announcement; identifies the internal version of Astra, the ten results, and OpenAI’s stated preparation/formalization process. https://openai.com/index/ten-advances-in-mathematics/
2. [2] OpenAI, “Ten Advances in Mathematics and Theoretical Computer Science,” PDF, updated 6 August 2026. Chapter 5, §6 “An affine specialization of the permanent” (especially PDF pp. 127–128); §12.3, “Related work” (PDF p. 148). https://cdn.openai.com/pdf/ten-proofs-oai.pdf
3. [3] Feinsilver and McSorley (2011), “Zeons, permanents and determinants,” International Journal of Combinatorics, DOI 10.1155/2011/539030. https://doi.org/10.1155/2011/539030
4. [4] Zenodo support, “What is a DOI?” explaining persistent record identifiers and DOI resolution. https://support.zenodo.org/help/en-gb/18-general/216-what-is-a-doi
5. [5] Zenodo help, “Search guide,” explaining public record-search and publication-date queries. https://zenodo.org/help/search
6. [6] Lewis et al. (2020), “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.” Establishes RAG as a documented architecture combining a pretrained model with retrieved non-parametric documents; used here to explain the proposed pathway, not as proof it occurred in this case. https://arxiv.org/abs/2005.11401
7. [7] OpenAI API documentation, “Web search.” Describes model toolchains that can search the web and return sourced online results during generation; cited as evidence that inference-time search is an operationally available class of mechanism, not evidence that this specific internal Astra run used it. https://developers.openai.com/api/docs/guides/tools-web-search

Record note: The 1 March 2026 date is author-identified. Attach the public Zenodo record URL, DOI, and metadata export; the supplied /me/uploads URL is account-private.